Data Cleaning

The raw eCommerce clickstream data was cleaned. The objective was to improve data quality, remove records that could distort checkout drop-off and Lost GMV analysis, and prepare the dataset for further analysis in MySQL and Power BI. Cleaning was done and prepared for analysis through the following steps:

Loaded Monthly Clickstream Dataset

Loaded the raw November 2019 eCommerce clickstream dataset containing 67M+ raw interaction records across 9 columns.

Removed Missing User Sessions

Identified and removed 10 records with missing user_session values (NaN), as valid session identifiers are strictly required to track user journeys across the conversion funnel.

Removed Duplicate Records

Identified and removed 100,519 duplicate records.

Converted Date-Time Column

Converted event_time from object format into proper datetime format to enable time-based analysis.

Handled Missing Categorical Values

Filled missing values in category_code and brand with 'Unknown' to ensure completeness in category and brand-level reporting.

Feature Engineering

To make the cleaned data more useful for analyzing product hierarchies and cart abandonment patterns across departments, several new fields were created by splitting category_code:

Main Category

Created main_category by extracting the top-level category (e.g., electronics, appliances, computers) from the first segment of category_code.

Sub Category

Created sub_category to capture mid-level product groups (e.g., kitchen, audio) for 3-part category codes, setting 2-part codes as 'Unknown'.

Product Type

Created product_type by extracting the specific item type (e.g., smartphone, tv, washer, notebook) from the final segment of category_code.

Data Storage & Ingestion

To enable high-performance querying and reporting in MySQL and Power BI, the finalized dataset was stored and ingested through the following steps:

Saved Cleaned Dataset

Exported the final cleaned dataset of 67M+ records and 12 columns to 2019-Nov-cleaned.parquet for high-compression, fast columnar access.

Loaded Dataset to MySQL Database

Streamed the cleaned data into the ecommerce_events table in MySQL (cart_abandonment database) using SQLAlchemy and PyMySQL with a chunk size of 100,000 records.